Advanced Control of Phase Space Presentation of Slow-Extraction Separatrix at ES

M. Fraser, B. Goddard, V.Kain, M. Pari, L. Stoel, F. Velotti

EDMS DOC **2280001 v.1**

https://edms.cern.ch/document/2280001/1

Slow extraction at the CERN SPS

- First slow extraction to SPS North Area:1978
  - Transfer lines designed for 400 GeV/c

- p^+: 400 GeV/c extraction momentum
- Heavy ions: ~ 30 GeV/c to ~ 380 GeV/c proton equivalent
  - Very low intensities

SPS parameters 2018 proton operation

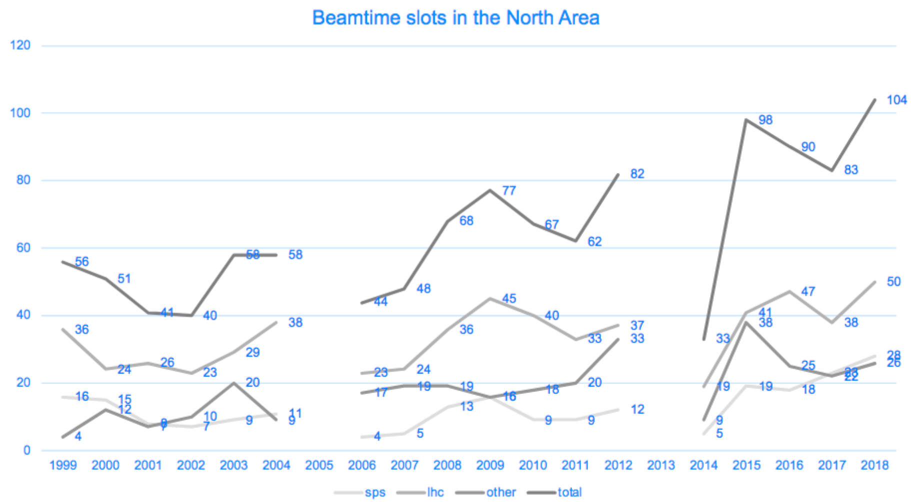
Extraction Momentum [GeV/c]	# p^+ at flattop [$\times 10^{13}$]	Min. repetition period [s]	flattop length [s]	# cycles/day
400	3.5	14.4	4.8	~ 4000

max. 4 x 10^{13}

min. 1.2 s

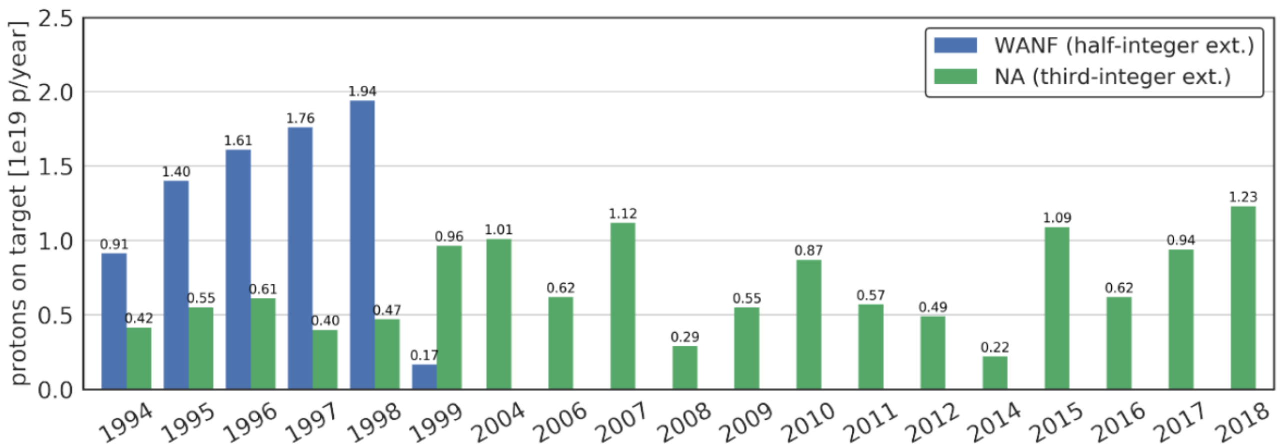
Evolution of beam requests over years

- Ever increasing interest in NA fixed target physics



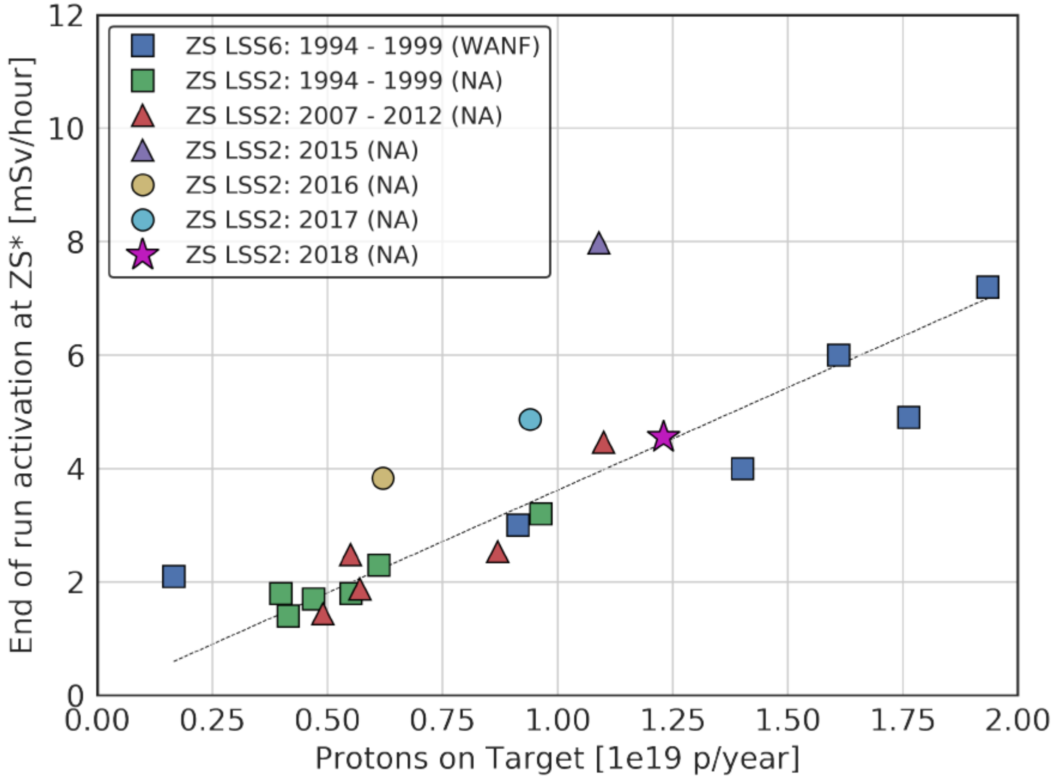
Courtesy K. Elsner
From *40 years NA physics Celebration*

- Record total intensity extracted to NA in 2018



Many new ideas for NA → "Physics Beyond Colliders"

- New projects → increase further the number of protons extracted to NA (e.g. with SHiP: 5e+19 p+/year)



* Measured ~30h after shutdown at ~1 m from beam line, peak at ZS plotted.

- **…and keep activation constant** (level of ~ 1e+19)

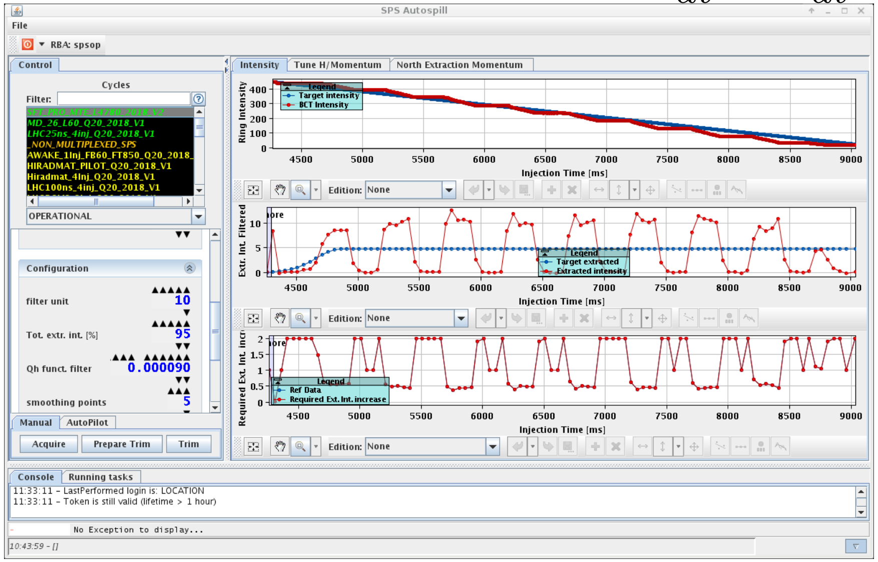
Ideas to reduce losses/activation in extraction channel

- Various ideas for loss reduction
  - New ES materials,…

- → slow extraction with **crystal shadowing** and **octupole phase-space folding;** but…
  - **Limited angular acceptance** for crystal shadowing
  - Requires **constant separatrix**/optics for octupole phase-space folding,…

Driving slow extraction at CERN SPS

- Third order resonance slow extraction **driven by tune sweep**
- Chromaticity $\xi_x = -1$, increase momentum spread through RF gymnastics to $dp/dt = \pm 0.15\%$
- Hardt-condition not fulfilled at the SPS, but D_x, D'_x small
- Feed-forward correction of Q_h function (main quadrupoles) in the SPS to linearize ring intensity decay on the slow extraction flattop
  - Correction algorithm built exploiting $\frac{dQ}{dt} \propto \frac{dI}{dt}$

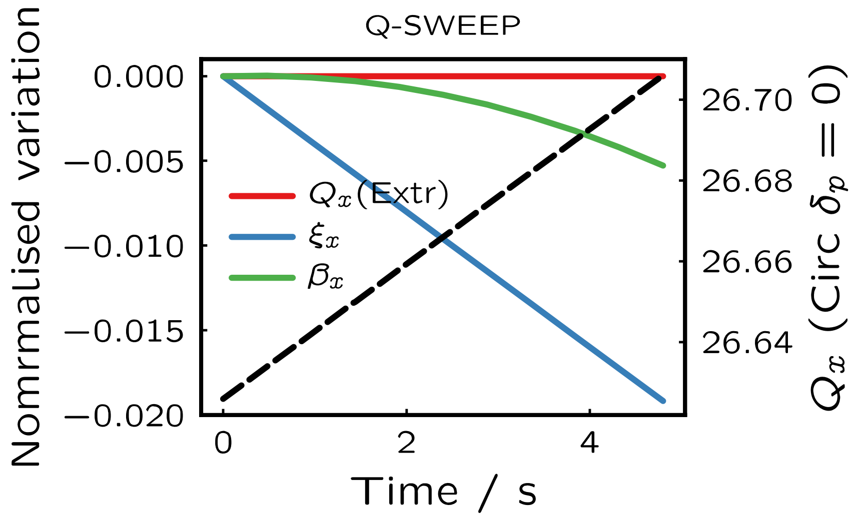


Autospill control room application

- No servo-quadrupole feedback system in the SPS since 2015

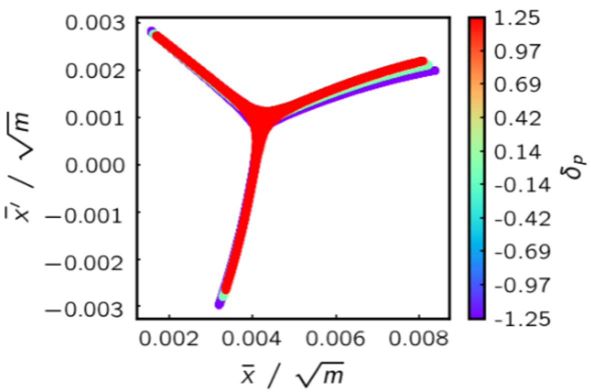
Evolution of separatrix during slow extraction

- Tune driven slow extraction → changing machine optics



β_x @ ES

- → rotating separatrix @ ES

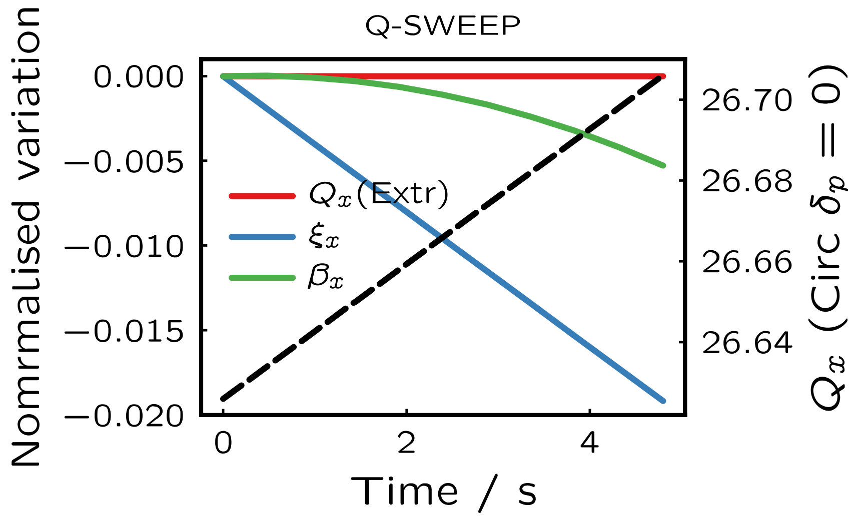


→ particles with different momenta arrive on different separatrices

Lowest momenta particles, extracted first.
Extraction momentum is proxy for time on extraction plateau.

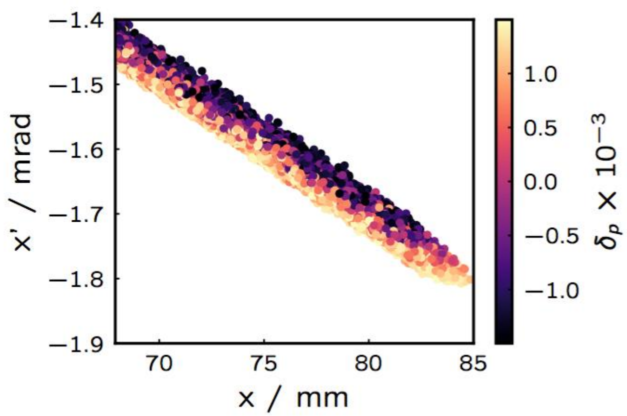
Evolution of separatrix during slow extraction

- Tune driven slow extraction → changing machine optics



β_x @ ES

- → rotating separatrix @ ES

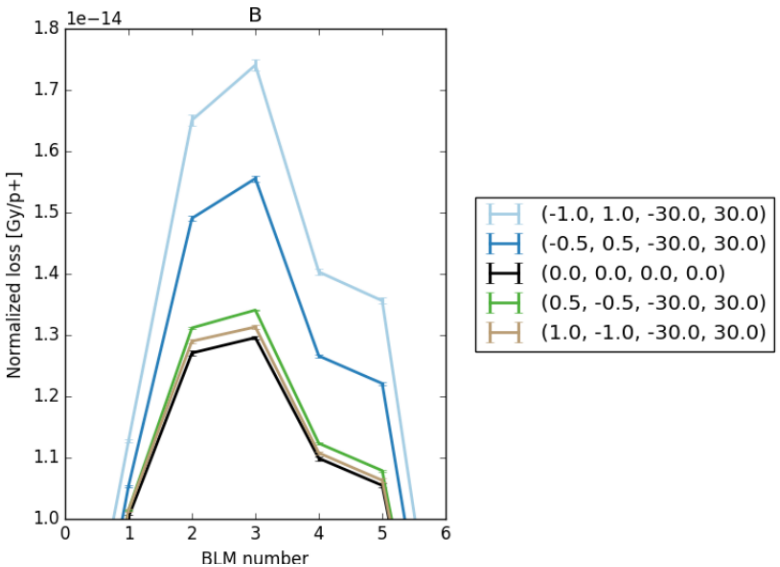


Phase-space distribution of entire spill of extracted beam @ ES.

Lowest momenta particles, extracted first.
Extraction momentum is proxy for time on extraction plateau.

Is it possible to compensate this?

- Use dynamic bump? Orthogonal knobs in x and x' at ES



Test with linear tune sweep, x and x' sweep.

Feed-down effects on tune to be compensated.

Implementation details were not finalized due to other idea…

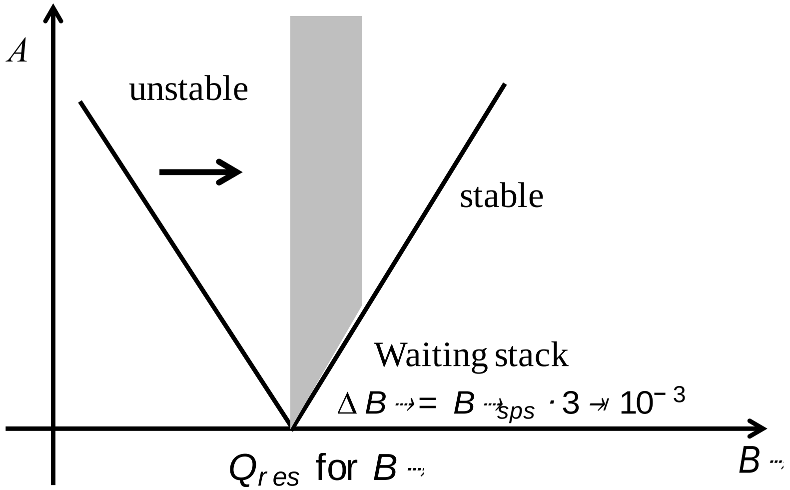
- Beginning of 2018 run: idea of running slow extraction betatron-core-like without beatron core
- → Constant optics slow extraction (COSE)

COSE Concept

- Separatrix change is consequence of moving resonance to select particles to be extracted via their momenta
- COSE: keep resonance fixed and scale machine settings according to momentum to be extracted.
  - → normalized strengths stay constant; fields change with momentum (or $B\rho$)

$$\frac{\delta^n B(t)}{\delta x^n} = k_n \times B\rho(t)$$

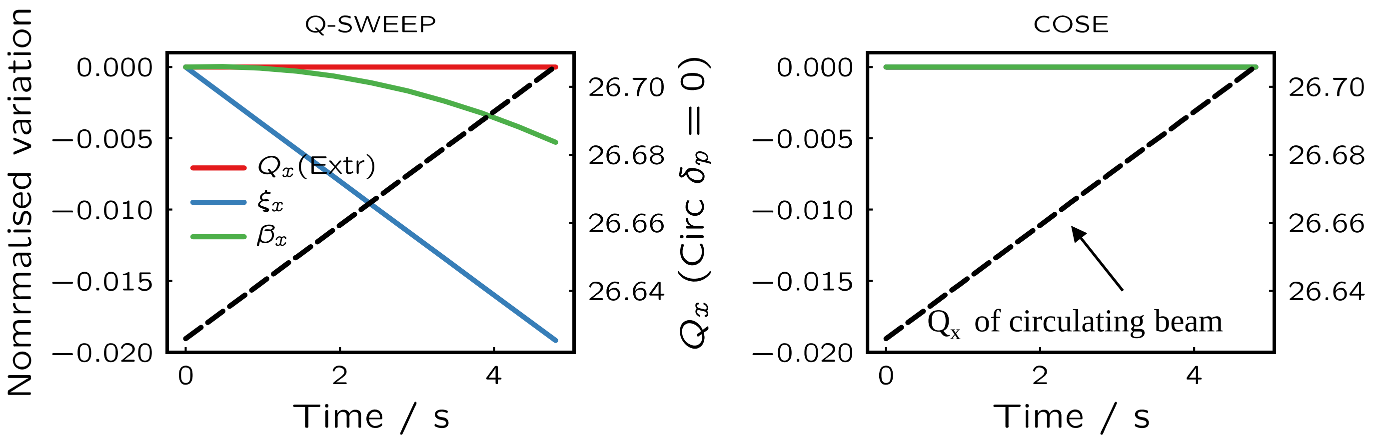
k_n: constant



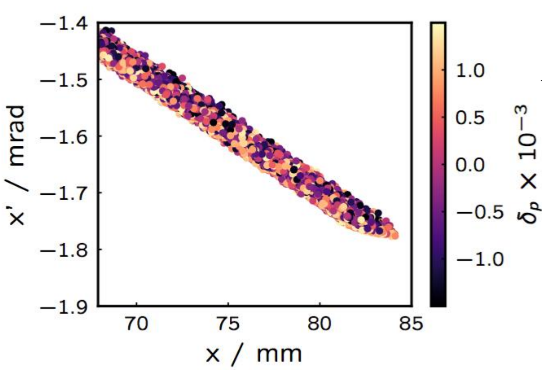
- In many ways similar to betatron core
  - Difference: momentum of extracted beam not constant

What would we expect from simulation?

- Optics is constant; circulating beam is off-momentum; extracted beam on-momentum; tune resonant for on-momentum beam

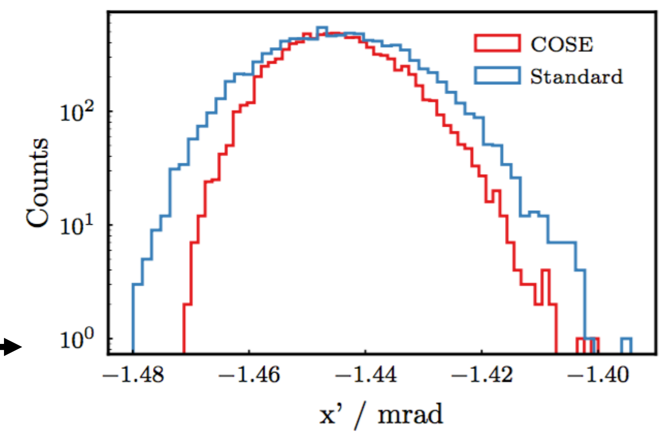


- Simulation of particle distribution at ES



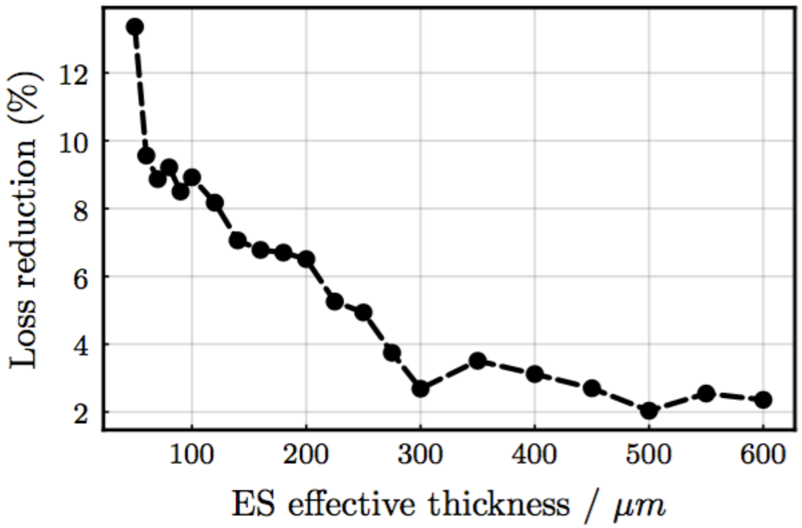
Correlation between x' and momentum disappears.

Reduction of angular spread by ~ 20 %.



Expect also loss improvement?

- Parametric study of dynamic bump in simulation



Loss reduction with reduction of angular spread?

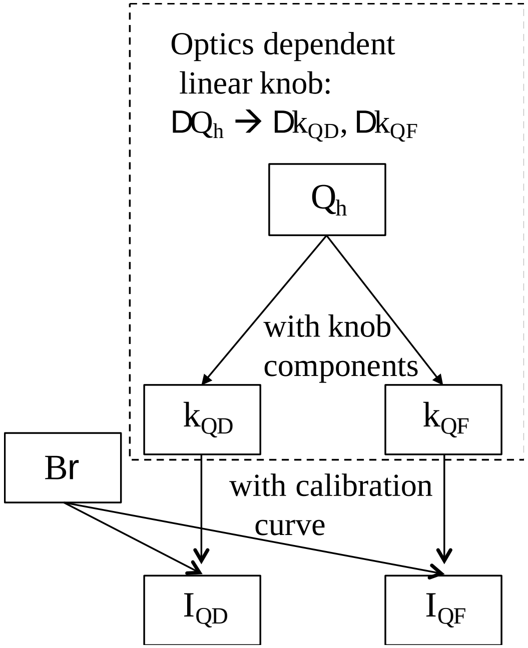
→ Depends on ES effective thickness

- Measurements from 2018 indicate ES effective thickness of ~ 500 μm
  - → very little loss reduction expected with COSE
  - → main purpose reduction of angular spread, fix optics and separatrix
  - → important for loss-reduction techniques like crystal shadowing

Implementation? → SPS settings management

- SPS control system based on LHC Software Architecture (LSA)

- "Only" work with normalized strengths k_n and high level physics parameters
  - E.g. physics parameter Q_h defined on top of k_{QD} and k_{QF}.
  - All functions of time in "cycle"
  - The fields/currents functions calculated
    - using centrally stored calibration curves
    - and $B\rho$ or momentum function
  - $B\rho$ or momentum function can also be trimmed
    - → re-calculate all fields/currents



- One trim of $B\rho$ in the SPS = update several 100s of parameters

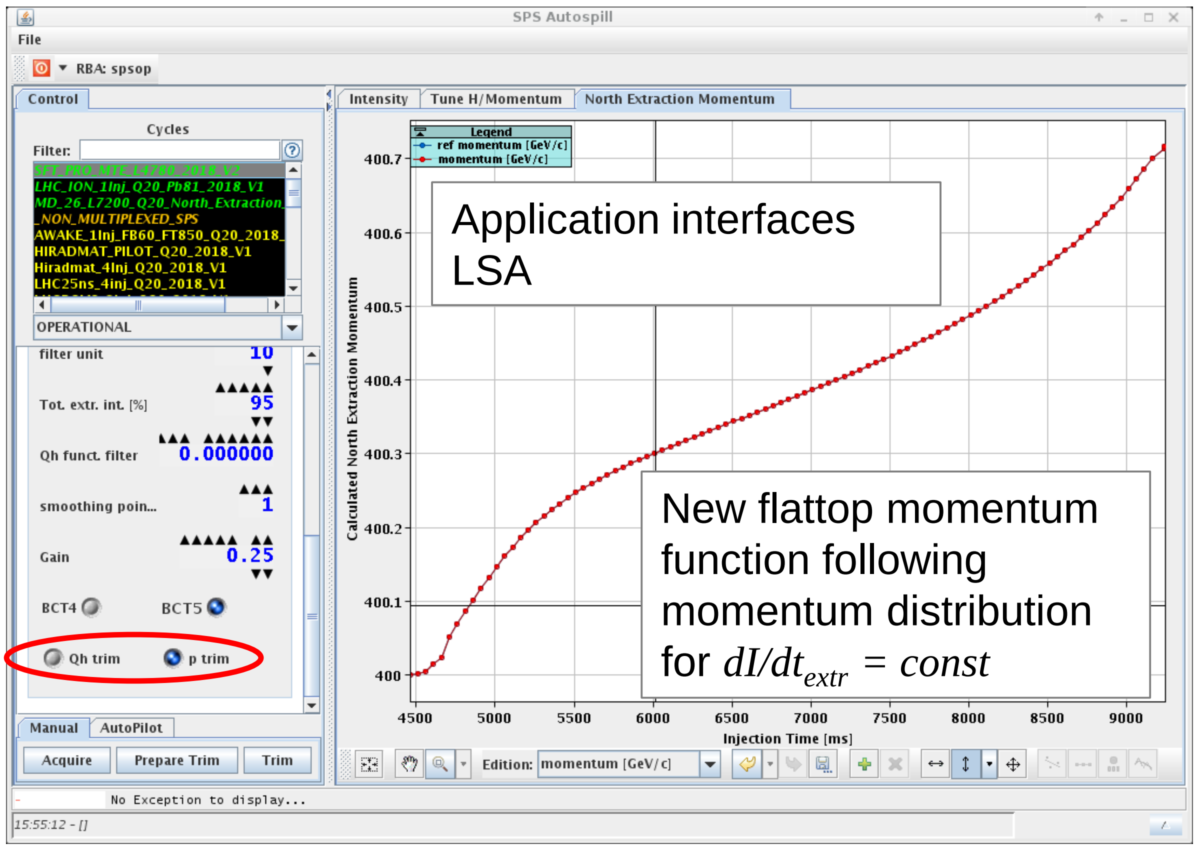
D. Jacquet et al., LSA - The High Level Application Software of the LHC and its Performance during the first 3 years of Operation, ICALEPS, San Francisco, CA, USA, 6 - 11 Oct 2013, pp.thppc058.

Implementation? → SPS settings management

- Already all prepared in *Autospill*…

$$\frac{dQ}{dt} \propto \frac{dI}{dt} \rightarrow \frac{dp}{dt} \propto \frac{dI}{dt}$$

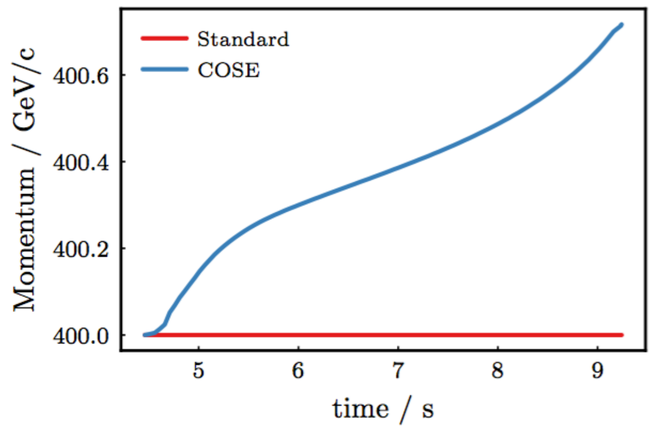
SPS works with momentum p instead of $B\rho$



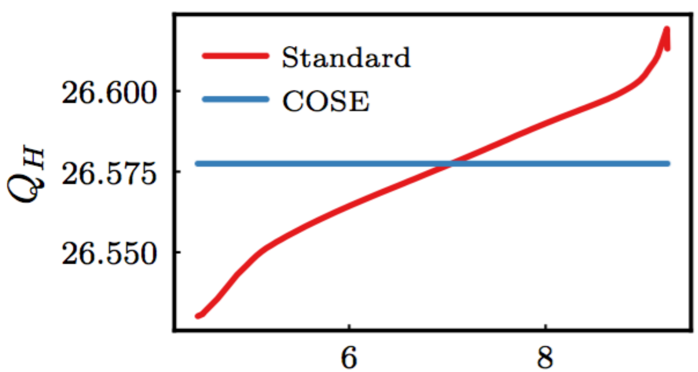
One open issue: transactional behavior. Will come post-LS2

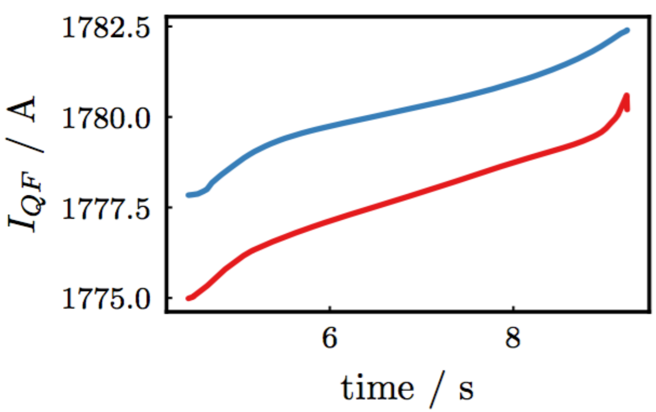
What does COSE look like?

- The slow extraction flattop momentum function follows the momentum distribution



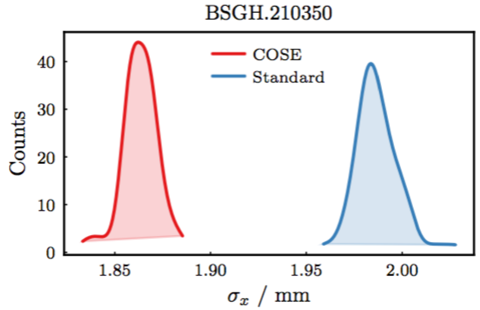
- All high level parameters stay constant





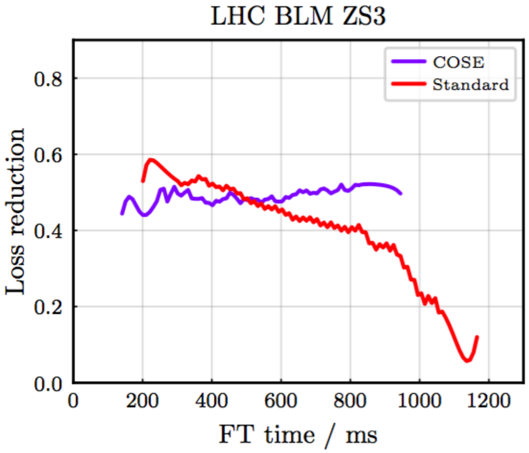
Confirm expectations from simulation in measurements?

- Reduced angular spread → reduced beam size of total spill in transfer line to NA
  - Expect 10 % beam size reduction at locations with SEM grids from 20 % reduction of angular spread



→ Confirmed by measurement

- Loss reduction experiment with bent crystals at ES
  - Most direct proof of constant separatrix through spill → constant loss reduction



Crystal aligned to shadow ES. Loss reduction as a function of time for 1.2 s flattop.

Setting up COSE

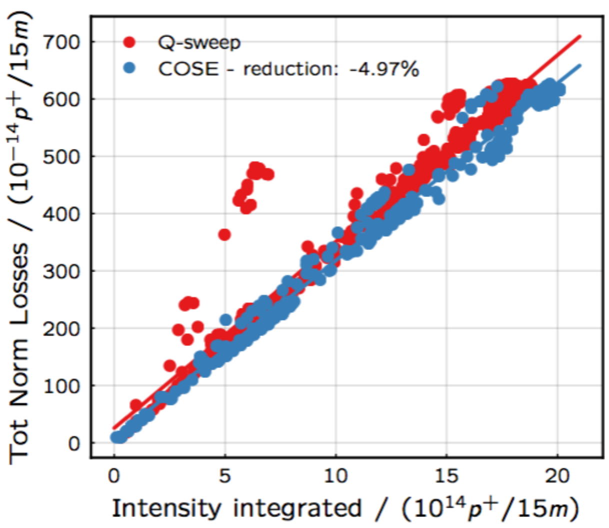
- Need to know momentum spread
  - E.g. can set up Q sweep extraction with Autospill, measure chromaticity → get momentum spread

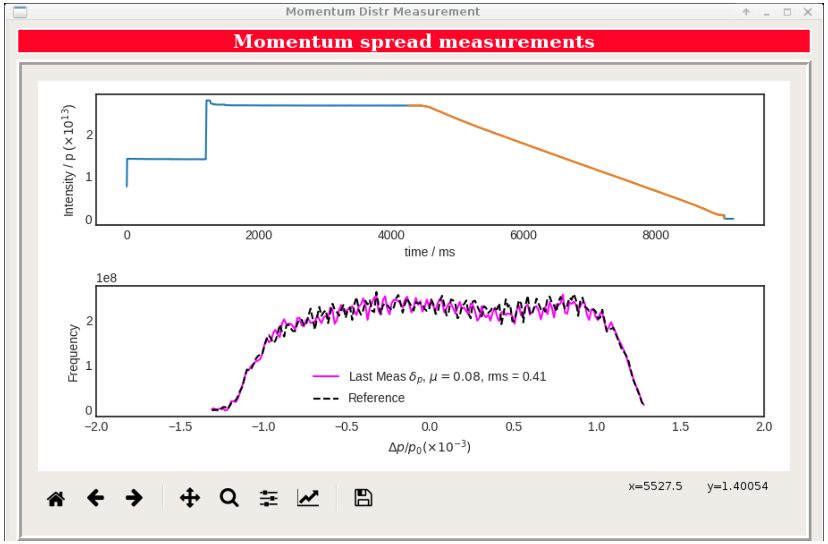
- Move beam radially by + ½ x $\Delta p/p$
  - → The lowest momentum beam should be on-momentum with the machine
- Sweep tune in Trim editor to find resonant setting
  - When slow extraction starts.

- Adjust tune constant at resonant setting along flattop
- Move last point of momentum to 400 GeV/c + $\Delta p/p$
- → Autospill

Operation with COSE

- COSE tested during machine development in summer 2018
- Put in operation on 20th of September, 2018 for the rest of the proton and ion FT run
- No degradation of spill ripple or other issues; losses at similar level





- Byproducts: how to adjust Brho in transfer line for stable trajectory, automatic "measurement" of momentum distribution

Summary

- Continuously growing demand for POT on the SPS North Area targets
- Activation needs to remain constant → losses reduction

- Loss reduction concepts require constant separatrix, reduced angular spread

- Constant optics slow extraction (COSE) removes the rotation of the separatrix over time by driving slow extraction with $B\rho$ at constant magnetic strengths
  - Removing the need of dynamic bump

- COSE straight forward implementation at the CERN SPS due to
  - High level parameter settings management based on LSA
  - Previous experience with feed-forward spill control

- COSE has become the new standard for driving slow extraction at the SPS